So you've decided to run a deep learning model on some data but are frustrated by the system itself? It's not a huge deal to the beginner but those who have a good idea of what tools are best for what sort of operations, usually have a less tiresome time running the system and training the entire model.

While Deep Learning may sound like a ridiculously power hungry technology, it really comes down to the context of the application. There's a plethora of Deep Learning hardware based on the complexity of the workload. Moreover, you require the powerful hardware for the training phase of an AI model, once that's done you can run inferencing on low-powered devices on the edge. So if you're starting out with Deep Learning, then your existing desktop computer is more than sufficient. However, specialised hardware does exist so let's understand where it comes into play.

IC Vendors	Intel, Qualcomm, Nvidia, Samsung, AMD, Xilinx, IBM, STMicroelectronics, NXP, MediaTek, HiSilicon, Rockchip	12
Tech Giants & HPC Vendors	Google, Amazon_AWS, Microsoft, Apple, Aliyun, Alibaba Group, Tencent Cloud, Baidu, Baidu Cloud, HUAWEI Cloud, Fujitsu, Nokia, Facebook	11
IP Vendors	ARM, Synopsys, Imagination, CEVA, Cadence, VeriSilicon, Videantis	7
Startups in China	Cambricon, Horizon Robotics, DeePhi, Bitmain, Chipintelli, Thinkforce	6
Startups Worldwide	Cerebras, Wave Computing, Graphcore, PEZY, KnuEdge, Tenstorrent, ThinCI, Koniku, Adapteva, Knowm, Mythic, Kalray, BrainChip, Almotive, DeepScale, Leepmind, Krtkl, NovuMind, REM, TERADEEP, DEEP VISION, Groq, KAIST DNPU, Kneron, Esperanto Technologies, Gyrfalcon Technology, SambaNova Systems, GreenWaves Technology	28

Some of the companies working on AI related hardware and software IPs

WHAT MAKES A GOOD DEEP LEARNING SETUP

The most common question you might have encountered when looking for the best tools to buy on the market might have been-"What makes one better than the other?" Are CPUs better than GPUs? Or is it the other way around? When it comes to CPUs or GPUs, what parameters matter the most?

First of all, understand that most operations in deep learning projects are bandwidth bound but there are cases that aren't as memory bound.